

Tilburg Substation EMC – Project overview and tracker

Project: Advisory services Tilburg Substation EMC

Client: SC&M II V.O.F. | **End Client:** TenneT

DNV Reference: 00431480-TDT 25-1641

Date : 12-12-2025

Meeting list:

1. Kickoff – Held on 5 December 2025, 09:30, Attendance (Peet, Juriaan)
2. 2nd Feedback meeting – [20 January 2026 Feedback meeting](#)

1 PROJECT OBJECTIVE

To conduct a Conductive influence evaluation for the new Tilburg substation project, focusing on the impact of short circuits on nearby 3rd party metallic objects (pipelines, data cables, fences, etc.) based on NEN 3654:2023 and EN 50522.

A critical objective is to address the specific challenge regarding high soil resistivity (approx. $500 \Omega \cdot m$) which currently results in a calculated station resistance (R_d) of 0.931Ω and a potential Ground Potential Rise (GPR) of up to 27.1 kV. The study aims to verify these values and determine if mitigation is required for nearby assets.

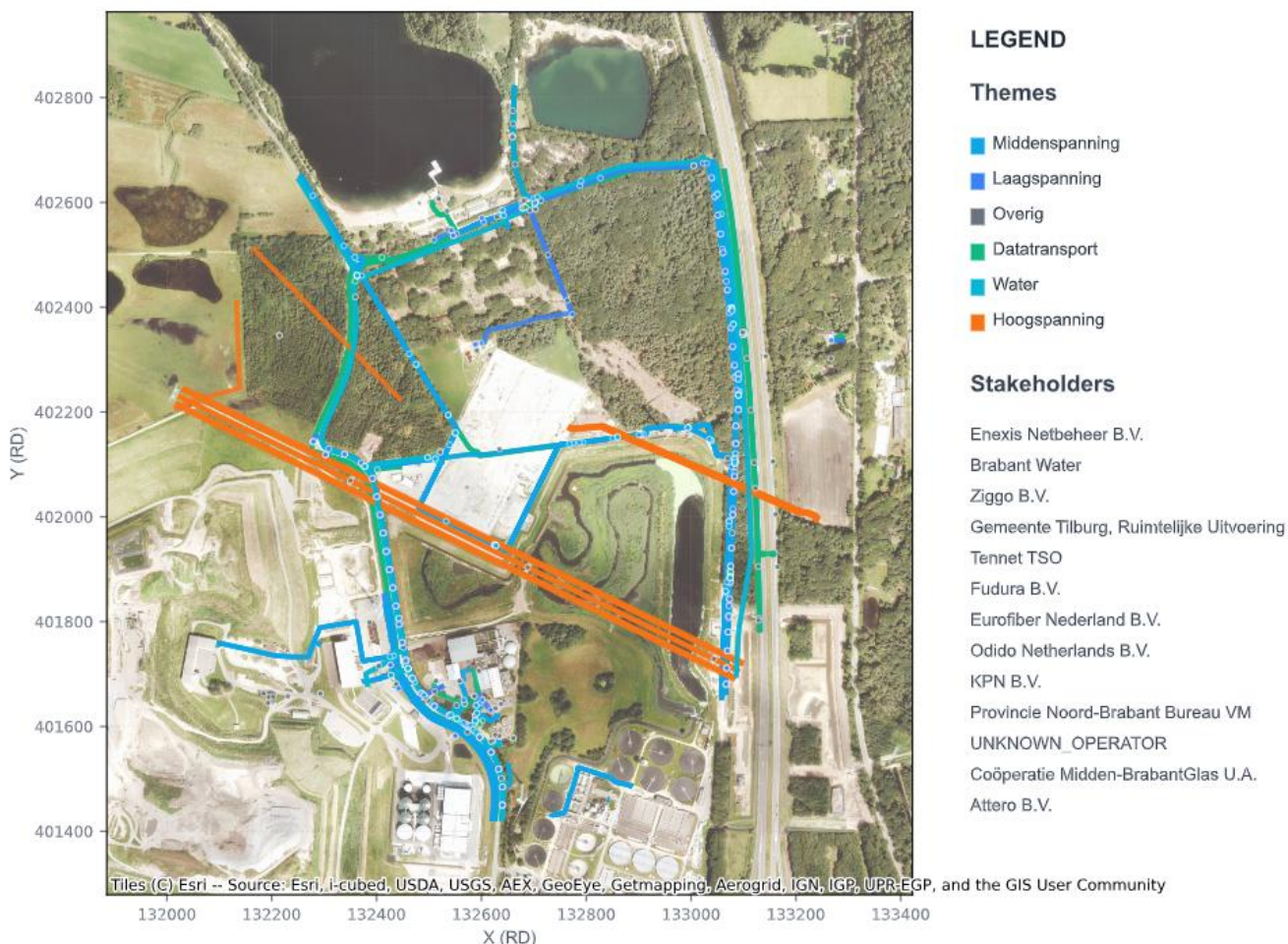


Figure 1: Tilburg substation and surrounding 3rd party assets under study

2 SCOPE OF WORK & METHODOLOGY

DNV will execute the project in the following phases:

Task 1: Analytical Modeling & Danger Zone Assessment

- **Activity:** Perform initial analytical modeling to determine the Ground Potential Rise (GPR) and identify the immediate danger zones.
- **Output:** High-level identification of assets falling within critical influence areas.

Task 2: Data Verification (KLIC vs. Reality)

- **Activity:** Verification of input data. Specifically, clarifying the status of the pipeline indicated on current KLIC data (which is believed to be removed) versus the actual site conditions.
- **Requirement:** Client to provide updated "Click data" or as-built drawings to ensure the model does not calculate interference for non-existent assets.

Task 3: Detailed Simulation (CDEGS/XGSLAB)

- **Activity:** Preparation of a detailed electrode model based on substation design inputs.
- **Simulation:** Calculation of voltage rise in the soil using a fault current of 29.1 kA.
- **Analysis:** Plotting results on the terrain against updated asset locations to check compliance with limits.

Task 4: Reporting

- **Activity:** detailed EMC influence report including calculations and mitigation recommendations (if applicable within the 16-hour scope).

3 KEY DELIVERABLES

DNV will deliver the following:

1. **EMC Influence Report:** A comprehensive report in English including detailed calculations, methodology, and results plotted against terrain data.

4 SCHEDULE & TIMELINE

The timeline is based on a 14-week duration as per the proposal.

Milestone	Activity	Baseline Date	Current Forecast	Status	Comments / Changes
T0	Project Start / Kick-off	05-12-2025	05-12-2025	CLOSED	Kick-off meeting scheduled.
M1	Data Collection / Verification	19-12-2025	19-12-2025	CLOSED	Critical: Confirmation of removed cable data.
M2	Draft Report Delivery	12-02-2026	12-02-2026	PLANNED	Based on 14-week schedule minus review time. It was discussed during the kickoff to aim for end of January Or beginning of February.
M3	Client Review Period	26-02-2026	26-02-2026	PLANNED	2 weeks allocated for stakeholder feedback.
M4	Final Delivery	12-03-2026	12-03-2026	PLANNED	Final version.

5 MINUTES OF MEETING & ACTION LOG

Project: Tilburg Substation EMC

ID	Category	Description	Owner	Due Date	Latest Status / Comments	Status
A01	Inputs	Asset Verification: Confirm removal of the pipeline running through the substation terrace. Current KLIC data is outdated. Provide updated drawings/data. MV cable on site relocated and running now along the fence	Client (SC&M) Juriaan Koppes	ASAP	[05-12]: Discussed in Kick-off preparation. Essential for accurate modeling.	Closed

6 DECISIONS & AGREEMENTS

ID	Category	Decision / Agreement	Date Agreed
D01	Team	DNV Project Manager: Peet Schutte. Engineers: Ivan Grobelaar, Asim Al sofi.	05-12-2025
D02	Method	Modeling Approach: The team will first model analytically to check the GPR danger zone before proceeding to detailed CDEGS simulations.	05-12-2025

7 RISK REGISTER (WATCHLIST)

ID	Risk Description	Potential Impact	Mitigation Strategy	Status
R01	High Soil Resistivity	Deep soil layer resistivity ($\sim 500 \Omega \cdot m$) may cause Station Resistance (R_a) to exceed the 0.1Ω target (currently calc. 0.931Ω), leading to high GPR (27.1 kV).	DNV to perform detailed CDEGS modeling to verify conservatism of current assumptions. Highlight specific mitigation needs early.	Active
R02	Input Data Accuracy	Modeling based on outdated KLIC data (e.g., phantom pipeline) will result in false positives for interference.	Client to verify "Click data" and confirm removal of assets before final simulation runs.	Active